

Two Models, One Compiler: Verifier-Centric Coordination of Two Fixed Off-the-Shelf LLMs for Competition Mathematics in Lean 4

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Preprint · Verified Mechanisms take-home submission · August 30, 2026

Abstract

Problem. Two fixed models — qwen/qwen3.5-flash-02-23 and openai/gpt-oss-120b — must jointly solve competition-mathematics problems and prove the results in Lean 4 under per-problem caps of \$1.00 and 8 hours, scored by a kernel-level Comparator. **Method.** We build a verifier-centric staged controller: deterministic tactics, pooled cross-model sampling, shallow compiler-guided repair, cross-model handoff on repair plateaus, sketch-and-fill decomposition, and a *comparator precheck* that validates accepted proofs against the actual scoring gate rather than the faster development checker. **Main controlled result.** At matched short budgets, the controller improves both solo models over the kit's raw baselines (7/16 and 8/16 raw; 8–9/16 each under the controller), and the two-model system attains slightly higher aggregate coverage (9–10/16) with clearly complementary per-problem failures: each family solves problems the other misses under every seed tested. The aggregate gap (+1) sits within our observed run-to-run variation, so we treat per-problem complementarity, not the aggregate, as the primary evidence. **Hard instances.** Multi-hour decomposition runs solved three of the four provable hard problems that no short-window configuration solved; a single-model arm given the same controller and a long horizon reproduced one of these, suggesting that verifier-aligned search infrastructure and search horizon account for more of the gain than model-to-model interaction, with two-family coverage and robustness as smaller, useful additions.

1 Introduction

Formal theorem proving allows a language-model system's output to be judged exactly: a Lean 4 proof either type-checks against the stated theorem or it does not. The Verified Mechanisms task instantiates this setting with two fixed, off-the-shelf models served through OpenRouter — Qwen 3.5 Flash (fast, inexpensive, February-2026 training cutoff) and GPT-OSS-120B (slower on the accessible provider tier, cheaper per token, strong at high reasoning effort, June-2024 cutoff) — and asks for a coordination layer that makes them jointly solve and prove competition mathematics, together with a scientific account of whether the collaboration helps.

We organize the paper around three research questions:

- **RQ1 (scaffold).** Does the coordination scaffold improve either model over the same model in the kit's raw repair-loop baseline?
- **RQ2 (pairing).** Holding the scaffold fixed, does using both models improve over the better single model — and if so, through what mechanism?
- **RQ3 (hard instances).** What enables solutions on the problems that no short-window configuration solves?

Our design premise is that this task is a coverage problem attached to an exact verifier: the verifier makes candidate selection free, which shifts value away from dialogue-style coordination (for which compute-matched evidence is weak; §7) and toward candidate diversity, a single consensus decision the verifier cannot check cheaply, and a second set of priors when the first model stalls. Contributions: (1) a staged, individually-ablatable controller (Fig. 1) including a comparator precheck that aligns search with the true scoring gate; (2) a matched-budget experimental design that separates scaffold effects from pairing effects, plus a 24-problem custom benchmark with reference

proofs and a held-out split; (3) results for RQ1–RQ3 (§5), including a per-problem complementarity analysis; and (4) an ablation and robustness study (§6), including a ten-mechanism screening experiment and an unplanned provider-outage natural experiment.

2 Task and evaluation protocol

Scoring. A problem is solved iff the official *Comparator* — a separate scoring program that rebuilds the submitted file from a cold start, checks it with the Lean kernel, and demands kernel-level equality between the submitted statements and the challenge statements under a three-axiom whitelist — accepts every required declaration; numeric answers must be literal decimals; per-problem spend must not exceed \$1.00; and the run must finish within the time cap (8 hours at judging).

The development checker differs from the scoring gate. During search, candidates are checked by a fast REPL that keeps Mathlib loaded (“warm”) and strips imports; the Comparator builds cold. The two can disagree: a proof the REPL accepts can exceed the Comparator’s build-time limit. This gap drives a central design element (§3.3) and a benchmark-validation step (§4).

Budget accounting is fail-closed. If an LLM call fails in a way whose cost cannot be established, the harness permanently stops all further spending for that problem and the problem scores zero even if a correct proof was saved earlier. The controller therefore runs inside conservative rails: a self-imposed deadline with margin, explicit per-call timeouts, checkpointing of every improvement, and a degraded mode that finalizes the best saved candidate without further LLM calls if spending is stopped.

Cost structure. Observed per-problem spend is \$0.01–\$0.10 against the \$1.00 cap even on 8-hour runs (maximum observed \$0.95); wall-clock, not money, is the binding resource, dominated by GPT-OSS latency (2–8 minutes per high-effort call [28]) and serial REPL checks.

Evaluation regimes. Results below come from distinct regimes; every result is labeled with its regime, and comparisons are made only within a regime unless stated.

Regime	Caps	Problem set	Used for
Raw kit baselines	≈20 min / \$1	kit 16	per-model floor for RQ1 (provided with the kit; caps differ slightly from ours, so a floor, not an exact control)
Matched short-budget arms	30 min / \$1, 2 seeds	kit 16	RQ1, RQ2: same controller, model set switched
Development evaluations	20 min / \$1	custom dev-16	variant selection and mechanism screening (§6)
Held-out checks	20 min / \$1	custom held-8	promotion checks only; never used for selection
Long-horizon runs	2–8 h / \$1	hard subsets	RQ3; includes three validation runs at exact judge settings

Table 1: Evaluation regimes. Every result in §5–§6 is labeled with its regime; comparisons are within-regime unless stated.

Reporting conventions. Repeated identical-configuration runs on the dev-16 vary by up to ± 2 problems, driven by three problems whose empirical pass rate across repeated runs is roughly 50%; we therefore report per-problem outcomes wherever a claim depends on them and treat aggregate differences within ± 2 as inconclusive. “Cumulative coverage” counts a problem as covered if any run in the stated regime solved it; it is a coverage statement, not a per-run success rate.

3 System

Figure 1 shows the controller: a single anytime escalation ladder whose stages are individually switchable (implementation: `submission/agent.py`, ~1,900 lines over the kit's services). The two models do not exchange messages; they interact through shared artifacts — candidate files, error histories, proof skeletons — at a small number of narrow interfaces: pooled sampling (S1), independent answer votes (S0.5), plateau handoff (S3), and alternating sketch duty within decomposition (S4).

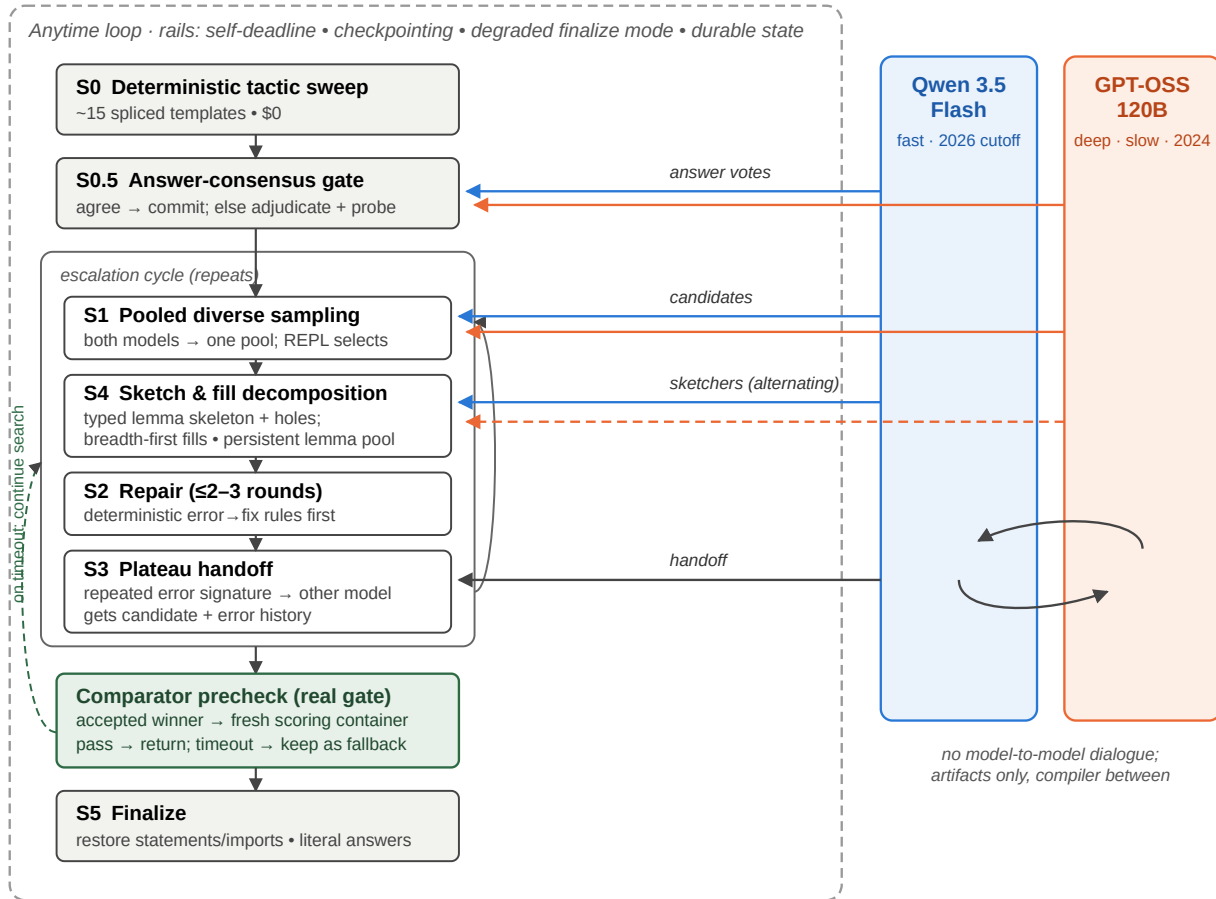


Figure 1: The controller. A staged escalation ladder (left) runs inside safety rails required by the fail-closed budget accounting; the two models (right) contribute through artifact-level interfaces only. Every REPL-accepted winner must pass a comparator precheck against the real scoring gate before being returned; a precheck timeout keeps the proof as a fallback while search continues.

3.1 Stages

S0, deterministic sweep (\$0). About 15 tactic templates, spliced into the pristine challenge statement, solve the easy third of the kit set before any model is called. Statement splicing — models produce proof bodies and helper lemmas, never statements — makes the Comparator's statement-equality requirement hold by construction. **S0.5, answer consensus.** For answer-type problems both models independently propose the numeric answer; agreement commits it, disagreement triggers one adjudication round plus inexpensive REPL falsification probes. This is the one decision the verifier cannot check cheaply, and a wrong commitment wastes unbounded proof search. **S1, pooled sampling.** Cheap Qwen samples, Qwen samples with reasoning mode enabled (an explicit thinking-token budget; hereafter *Qwen-reasoning*), and GPT-OSS samples enter one deduplicated pool; the REPL selects. **S2, capped repair.** At most 2–3 error-informed rounds with the model that wrote the candidate; deterministic error → fix rewrites (e.g., known option insertions, misnamed-constant substitution) are applied first, at no model cost. **S3, plateau**

handoff. A repeating error fingerprint routes the candidate plus a condensed error history to the other model for a fresh attempt, motivated by the models' complementary failure modes on the kit baselines (GPT-OSS repeats an identical failed attempt at low temperature; Qwen produces high-variance attempts that do not converge) and by their two-year training-cutoff gap. **S4, decomposition.** A sketcher (Qwen-reasoning primary, GPT-OSS alternating) writes an informal solution and a Lean skeleton of standalone, explicitly-typed helper lemmas with sorry placeholders; invalid skeleton lines are masked rather than discarded [11]; holes are filled breadth-first — every hole receives a cheap attempt before any hole receives multi-round repair calls — and proven lemmas persist in a per-problem pool across re-sketches and restarts. **S5, finalize.** Restore pristine statements and imports, enforce literal answers, re-verify, return.

3.2 Window-aware time constants

All stage and model time constants scale with the available window: a fixed 960-second GPT-OSS timeout cannot fit a 20-minute evaluation window and had silently disabled decomposition at short caps until scaled. The outer loop's cycle cap similarly yields to remaining time; an early version calibrated for slow mixed-model cycles stranded three hours of an eight-hour window when a provider outage made all-Qwen cycles fast (§6.3).

3.3 Comparator precheck

REPL-accepted proofs repeatedly exceeded the Comparator's cold-build time limit — three times on one problem, then on others — because kernel-side cost is invisible to the warm checker. The controller therefore validates accepted winners against the actual Comparator in a fresh scoring container before returning; a precheck timeout demotes the proof to a retained fallback and search continues for a cheaper proof. Section 6.2 quantifies the effect of this component; it involves no second model and is a pure verifier-alignment mechanism.

4 Experimental design

Arms. solo-qwen, solo-gptoss, and duo run the identical controller with only the model set switched, at identical caps and seeds, isolating pairing effects (RQ2) from scaffold effects (RQ1). The kit's raw baselines supply the scaffold-free floor per model.

Datasets and development boundary. The kit's 16 public problems informed development throughout (failure analysis, calibration); results on them are development-regime results except where the arms protocol makes the comparison internally controlled. To limit overfitting we authored 24 additional problems in the kit's format (number theory, modular arithmetic, divisor counting, factorial valuations, inequalities, telescoping products): 16 form a development set used for variant selection, and 8 were held out and used only to check promotions (two checks during the project; never used for selection). Three deterministic tactic-cascade entries were mined from archived failed subgoals of earlier runs and are therefore development-derived; they are generic tactic patterns, not problem-specific. The agent never observes reference proofs.

Benchmark validation. Each custom problem was admitted only after a reference proof was machine-checked in the development REPL. Because this paper's own argument is that REPL acceptance does not imply Comparator acceptance, all 24 reference proofs were additionally run through the full Comparator gate (fresh scoring container, manifest declaration names, cold build) before submission; the validation script and per-problem results ship with the repository (scripts/validate_references.py, outputs/reference-comparator-validation.json). All 24 reference proofs passed the Comparator (cold-build times 33–225 s, none near the gate's time limit), so the benchmark's solvability claims hold under the same gate that scores submissions.

Metrics and noise. Primary metric: problems solved under the stated regime's caps, with per-problem outcome grids for any claim that depends on composition. Aggregate differences within the observed ± 2 dev-16 band are reported as inconclusive. Where attribution matters (which stage or model produced the winning proof), we use the per-problem event logs shipped with each run.

5 Results

5.1 RQ1: the scaffold improves both solo models (matched short-budget regime)

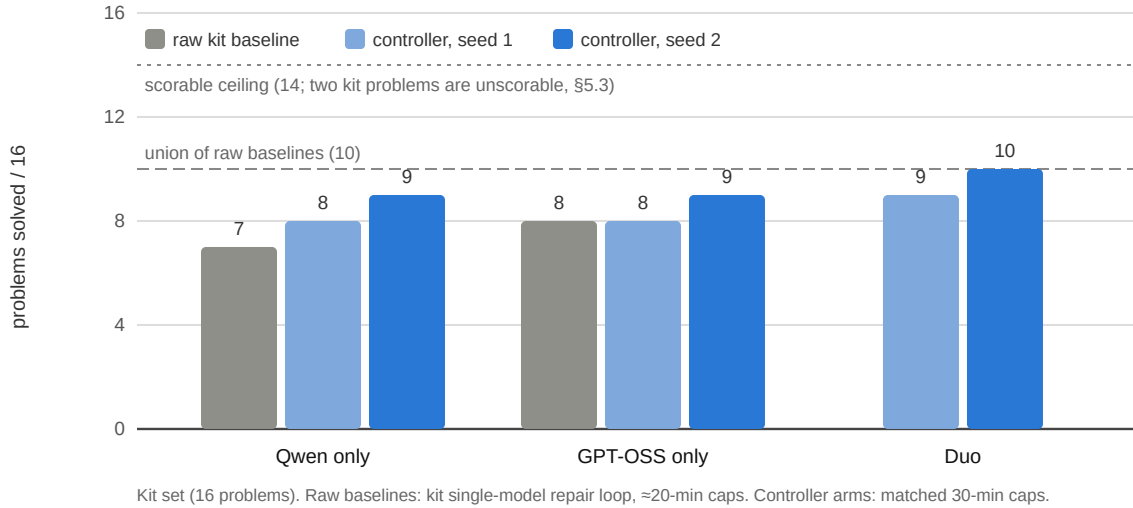


Figure 2: Matched short-budget regime. Same controller, model set switched; two seeds per arm; raw kit baselines shown as per-model floors.

Under the controller, each solo model matches or exceeds its raw baseline on both seeds (Qwen 7 → 8, 9; GPT-OSS 8 → 8, 9), with the free S0 sweep alone covering 6/16 before any model call. The raw baselines ran at slightly shorter caps, so we read Figure 2 as a floor comparison; the direction is consistent across both models and both seeds.

5.2 RQ2: pairing adds coverage through complementary failures, and the aggregate gain is small

The duo scored one problem above the best solo arm on both seeds (9 vs. 8; 10 vs. 9) at one-half to one-third of solo-Qwen's cost per run (\$0.16–0.17 vs. \$0.40–0.45), and in each seed matched the union of that seed's two solo arms. Because the aggregate difference sits within the ± 2 run-to-run band we observed on repeated identical configurations, we do not treat the +1 as strong evidence by itself. The per-problem grid (Table 2) is the more informative result: the two families fail on *different* problems, consistently across seeds.

Problem	Qwen s1/s2	GPT- OSS s1/s2	Duo s1/s2	Note
7 easy/medium problems	✓✓	✓✓	✓✓	solved by every arm, both seeds (6 of them by the \$0 sweep)
p07 (least divisible)	✓✓	✗✗	✓✓	GPT-OSS's answer gate agreed with itself on a wrong path both seeds
putnam_2018_a1	✗✗	✓✓	✓✓	solved in the duo by its Qwen sampling wave — the same wave that failed in the solo-Qwen arm, indicating sampling variance rather than a stable family difference
putnam_2020_a2	✗✗	✗✓	✗✓	GPT-OSS-side solve in seed 2 for both arms that carry it
p10 (factorial powers)	✗✓	✗✗	✗✗	solo-Qwen seed 2 solved it; the duo's seed-2 attempt was lost to a GPT-OSS-channel provider error (later cured upstream) — an honest cost of carrying the second channel

5 hard/unscorable problems	XX	XX	XX	out of reach for every arm at 30 minutes (see §5.3)
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Table 2: Per-problem outcomes, matched short-budget regime (✓ = Comparator pass; s1/s2 = seeds). The disjoint failures (p07 vs. the Putnam problems) are stable across seeds; the duo inherits both sides. p10 shows the regime’s honest counterexample.

Two further RQ2 observations from other regimes, labeled as such. *Robustness (unplanned natural experiment, long-horizon regime)*: during final validation the GPT-OSS provider refused requests for roughly 14 consecutive hours; a solo-GPT-OSS configuration would have scored zero over that span, while the duo degraded to Qwen-only behavior and continued solving. *Skeleton diversity (long-horizon regime)*: the first full-cap solve of p09 grew from a GPT-OSS sketch that Qwen repaired and filled — a proof structure the solo-Qwen arm did not produce in that run. We report this as a single observed instance, not a systematic effect.

5.3 RQ3: hard instances are governed by search horizon under decomposition

Six kit problems were unsolved by both raw baselines. Writing reference proofs for them established that only four are provable as stated: one is mathematically false as formalized, and one cannot elaborate under the Comparator’s true imports — a flaw invisible to the import-stripping REPL — so the scorable ceiling is 14/16 (certificates in the repository). Table 3 summarizes the record on the provable four.

Problem	Best outcome	Regime and evidence
p09 (IMO 1964)	Comparator pass; 3 of 3 full-cap runs	8 h/\$1 judge settings; \$0.10–0.91, 2–7.6 h; each pass required the precheck to reject a costlier accepted proof first
p10 (factorial powers)	Comparator pass	first solved at 2 h caps (\$0.05); solved again under judge-compatible checking (\$0.029, 38 min) after the precheck landed
rmo_2000_2	Comparator pass	8 h/\$1; \$0.75; 7.4 h of accumulated lemma-building in one uninterrupted window
rmo_2001_2	unsolved	best attempt (4 h, decomposition with skeleton persistence) ended one unfilled hole short

Table 3: Hard-tier record (cumulative coverage over the long-horizon regime, duo arm unless noted). Cumulative coverage across all regimes: 13 of 14 scorable kit problems. Both raw baselines: 0 on this tier.

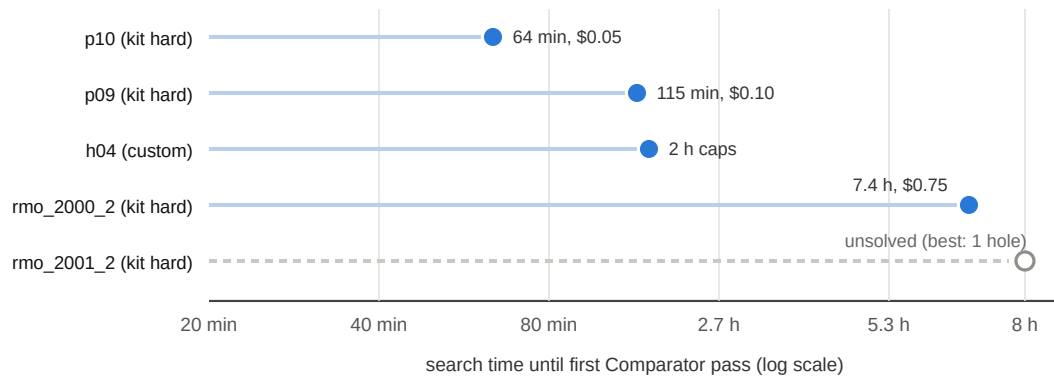


Figure 3: Time until each hard problem first passed the Comparator (long-horizon regime; hollow marker = still unsolved). For context, the dev-16 aggregate did not change when the short window was doubled from 20 to 40 minutes (scores 11–13 → 11, within the noise band): within the variants we tested, added short-window iteration did not solve these structural instances, while every instance that was solved fell at a multi-hour horizon under S4 decomposition. h04 is a custom problem that had not been solved by any of ~19 short-window configurations before the 2-hour run.

Two attribution results temper the collaboration reading of Table 3. First, a *solo-Qwen* arm given the same controller and a 4-hour window also solved p09, through the same trajectory (a costly accepted proof rejected by the precheck, then a cheaper fill that passed) — so on that problem the scaffold and horizon, not the pairing, carry the outcome. Second, the h04 solve's winning proof had itself been precheck-rejected — the precheck comparator timed out under concurrent machine load — and survived only because rejected winners are retained as fallbacks; on a loaded machine a precheck timeout is not reliable evidence that a proof is genuinely too costly. We note the 4-hour solo control is shorter than the duo's 8-hour runs, so it bounds rather than matches them.

6 Ablations and robustness

6.1 Mechanism screening on the development set

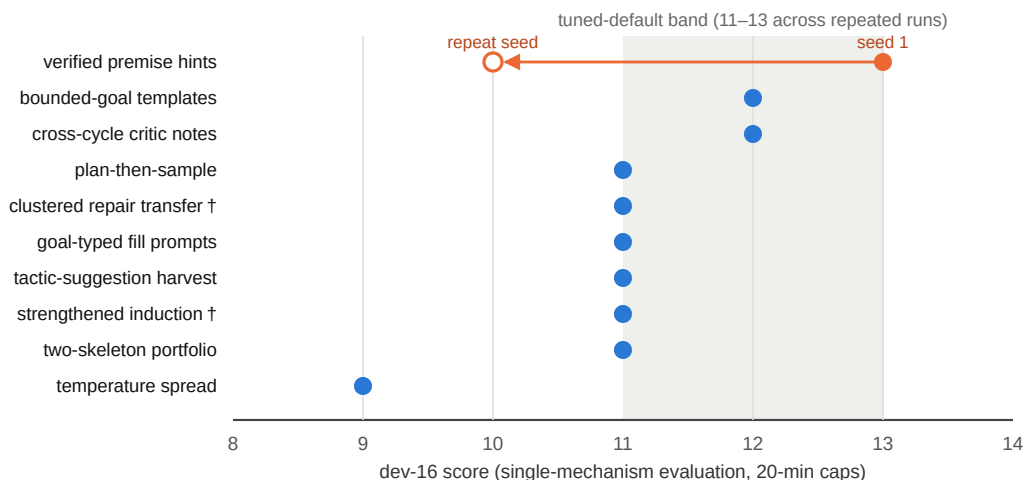


Figure 4: Mechanism screening (development regime). Ten mechanisms, each behind an independent flag, each evaluated with only that flag enabled. † = the mechanism's trigger condition fired at most once during its run, so its score is uninformative about the mechanism itself.

An initial selection loop promoted two changes (window-proportional time constants and breadth-first hole filling), raising the dev-16 from 11 to 13 with the held-out set stable at 6/8 on both checks. We then screened ten further mechanisms drawn from the literature and from our own failure analysis (Fig. 4). None of the ten produced a reproducible improvement over the tuned default under this evaluation: the only score above the default band (verified premise hints, 13/16, with every $\approx 50\%$ -pass-rate problem landing favorably) fell to 10/16 on its repeat seed, and two mechanisms' trigger conditions fired at most once. This screening is single-toggle and short-window: it does not explore interactions among mechanisms, and it says nothing about these ideas in other systems or at other horizons. A consistent pattern within it: every mechanism that lengthened prompts with guidance material scored at or below the default while costing more, suggesting that at short windows this system benefits more from additional cycles and sample diversity than from added instruction.

6.2 Precheck and reliability

The comparator precheck's contribution is directly measurable from run logs: one full-cap validation run scores 0/4 without it (two accepted-but-costly proofs would have been returned and rejected at scoring; with the precheck, both were rejected internally and a later proof passed); a problem that had been lost three consecutive times to scoring-time build timeouts was solved on the first post-precheck attempt; and the h04 fallback case (§5.3) shows the retain-don't-discard half of the design mattering in the opposite direction. Across the final configuration's runs — including container restarts that severed all in-flight connections, the 14-hour provider outage, and out-of-memory REPL

container failures — no run ended with unaccounted cost, and each of the twelve full-cap problem attempts is traceable in the shipped event logs to either a scored outcome or a specific, documented environmental cause.

6.3 Negative results retained

Deeper per-hole reasoning budgets at short windows: no additional structural solves, +16% cost. GPT-OSS inside short-window sampling waves: its call latency halves the number of completed cycles. The cycle-cap interaction of §3.2 was discovered when a provider outage made cycles fast and an 8-hour window ended at 4.9 hours; the fix (yield to remaining time) is in the submitted configuration. These are recorded, with run identifiers, in the repository's experiment log.

7 Related work

Repeated sampling against a verifier converts coverage into accuracy [1], and is the baseline any richer mechanism must beat at matched cost. Cross-family candidate pooling captures most ensemble benefit with two models [2], and pays when the selector is strong [3]; our selector is exact. Compiler-feedback repair helps most when shallow [4, 5, 6, 7]; equal-budget self-repair can lose to resampling [8], and models correct located errors better than they locate them [9] — which motivates handing a candidate plus located errors to a *different* model rather than critiquing. Sketch-then-fill decomposition underlies the strongest current systems [10, 11, 12, 13, 14, 15], and agentic loops around unmodified general models are now competitive with fine-tuned provers [7, 23, 24]; GPT-OSS-120B itself appears as the informal reasoner of an open hard-mode pipeline [25] and among the most cost-efficient provers in an independent evaluation [26]. Verifier-gated cascades [16] and self-consistency [17] supply the cost-allocation and answer-commitment patterns we adopt. We did not build dialogue-style coordination: compute-matched evaluations of multi-agent debate are largely negative [18, 19, 20, 21], and mixture-of-agents synthesis underperforms selection [22, 3]. Our evaluation methodology follows [27].

8 Limitations

Sample sizes are small (16 kit problems, 24 custom), and the dev-16 varies by ± 2 across identical runs; we therefore avoid significance claims and report per-problem outcomes. The RQ2 aggregate difference (+1 on both seeds) is within that band; the complementarity pattern of Table 2 is the sturdier finding. The mechanism screening is single-toggle and cannot exclude beneficial interactions among mechanisms. The long-horizon solo control (4 h) is shorter than the duo's full-cap runs (8 h). The raw baselines ran at slightly shorter caps than the matched arms. Development-environment conditions (provider outages, container restarts, shared-machine memory pressure) affected several runs; each incident is documented per-run, and the one development-only mitigation is inactive under judging defaults. Finally, these are two fixed mid-scale models: the demonstrated range — the full easy/medium tier plus several hard instances, with olympiad-final-tier problems unsolved — is consistent with published results for comparable backbones [6, 14, 25, 26].

9 Conclusion

When proof correctness is externally verifiable, most of the value of a multi-LLM theorem-proving system in our setting came from verifier-aligned search infrastructure and complementary candidate coverage rather than from conversational coordination. The scaffold produced the largest, most repeatable gains — it improved both solo models, and its comparator precheck decided multiple runs on its own. Using two model families added smaller but real benefits: per-problem coverage the aggregate score understates, one useful consensus decision, one observed cross-family proof structure, and robustness to a provider outage that would have zeroed a single-model configuration. Hard instances were governed by decomposition and search horizon: every hard problem that fell did so at a multi-hour horizon, one fell to a solo arm under the same scaffold, and none fell to short-window prompting variants. Within the mechanisms and budgets we tested, that ordering — scaffold, then coverage, then horizon; dialogue nowhere — is the paper's main empirical claim.

Reproducibility and disclosure. The repository contains the controller, the custom benchmark with reference proofs and Comparator validation results, per-run artifacts and event logs for every result above, and step-by-step reproduction instructions (RUNBOOK.md); `scripts/judge_check.sh` passes on the submitted configuration. Appendices: RESEARCH.md (design analysis), PART2.md, EXPERIMENTS.md, RESEARCH_LOOP.md. This submission was developed with substantial assistance from Claude (Anthropic) — code, experiments, analysis, and writing — as disclosed in the submission form.

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